Athanasios **Pavlides**

Computer Scientist

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"The best thing about a boolean is even if you are wrong, you are only off by a bit."

Summary

Computer Science graduate (BSc, University of Nicosia) specialising in cybersecurity, with hands-on industry experience as a Cybersecurity Analyst. Skilled in SIEM operations (Wazuh), intrusion detection (Suricata), incident response, and threat automation. Completed a Final Year Project integrating ML-KEM-768 (CRYSTALS-Kyber) post-quantum cryptography into Signal Desktop, graded A.

Education

AAU Klagenfurt — *Austria*

MSc in AI & Cybersecurity (MAICS) — *Starting Winter 2026/27*

UNIC (University of Nicosia) — *Nicosia, Cyprus*

B.S. in Computer Science, Minor in Mathematics — *Sep 2022 - Jun 2026*

- Specialisation in Cybersecurity, Algorithms, and Distributed Systems
- Relevant Coursework: Network Security, Assembly, Discrete Mathematics, Linear Algebra, System Analysis & Design, Data Communication, Calculus, Digital Systems
- Final Year Project: Post-Quantum Cryptography Integration in Signal Desktop using ML-KEM-768 (CRYSTALS-Kyber) — supervised by Prof. Harald Gjermundrød
- Co-authored paper submitted to MDPI journal on post-quantum cryptography (2026).

Skills

Security & SIEM : Wazuh, Suricata, IDS/IPS, MITRE ATT&CK, Threat Modelling, Penetration Testing

Cryptography : ML-KEM-768 (CRYSTALS-Kyber), Signal Protocol, Post-Quantum Cryptography, PKI

Programming : Python, TypeScript, C++, Java, Bash, SQL

Infrastructure : Docker, Linux, Git, REST APIs, Cloud VMs

Languages : English (fluent), Greek (native)

Programming Projects

Integrating Post-Quantum Cryptography into Signal Desktop — *Cyprus*

TypeScript, ML-KEM-768, Signal Protocol — *2025 - 2026*

- IEEE Full Paper accepted at CSNet'26 (10th Cyber Security in Networking Conference). Co-authored with Harald Gjermundrød & Ioanna Dionysiou, University of Nicosia. Paper: `https://edas.info/showPaper.php?m=1571301345`
- Implemented ML-KEM-768 (NIST post-quantum standard) as a hybrid quantum-resistant layer inside Signal Desktop — wrapping message content before Signal's Double Ratchet encrypts it, achieving NIST Level 3 security (AES-192 equivalent) with no protocol changes.
- Developed PQWrapper.ts TypeScript module integrating ML-KEM-768 into Signal Desktop's existing encryption stack using `@noble/post-quantum` and `@noble/ciphers`.
- Journal version submitted to MDPI (under review). Source code: `https://github.com/Somthin2/Signal-PQC-Integration`

Suricata Alert Automation System — *Cyprus*

Python, Bash, Webhooks, Suricata — *Mar. 2025*

- Engineered a real-time threat notification system integrating Suricata IDS with Telegram and Mattermost via Python and Bash webhooks.
- Implemented custom Active Responses for critical security events, significantly reducing manual log analysis overhead.

Custom Wazuh SIEM Monitoring Module — *Cyprus*

Python, JSON, Wazuh API — *Feb. 2025*

- Extended Wazuh's core monitoring capabilities with a custom Python module leveraging the Wazuh REST API.
- Automated log ingestion, threat correlation, and multi-channel alerting; improved detection accuracy through rule fine-tuning.

Social Connections Website

Cyprus

Java, Python, HTML, NoSQL, JSON, Flask

Sep. 2023 - Nov. 2023

- Full-stack web platform supporting real-time user communication, authentication, and session management.
- Source code available at: `https://github.com/Somthin2/Social-Connections-Website/tree/main#readme`

Multi-Floor Hospital Management Application

Cyprus

Java, MySQL

Dec. 2023

- Designed and developed a hospital management system handling personnel, inventory, and patient appointment scheduling across multiple floors.
- Source code available at: `https://github.com/Somthin2/AthanasiosMultifloorHospitalenduserHandlingSystem/tree/main#readme`

Experience

Catalink

Cyprus

Cybersecurity Analyst

Jun 2024 - Present

- Deployed and configured IDS/IPS solutions (Suricata, Wazuh) across private and public networks, monitoring 40+ agents for real-time threat detection.
- Maintained and upgraded SIEM infrastructure, including a Suricata 7.x to 8.x migration and OpenSearch cluster recovery from RED state (450+ unassigned shards) with zero data loss.
- Investigated and resolved a long-standing detection gap where 600+ logged malicious IPs were never blocked, correcting the automated block-list pipeline.
- Developed custom Active Response mechanisms in Python and Bash, automating threat mitigation and reducing incident response time.
- Built webhook-based alert pipelines integrating Wazuh with Telegram and Mattermost for real-time SOC notifications.
- Fine-tuned Wazuh SIEM detection rules to minimise false positives and improve detection accuracy across a mixed environment.
- Configured email security infrastructure implementing SPF, DKIM, and DMARC authentication protocols to protect against phishing and email spoofing.
- Designed and deployed GoPhish phishing simulation campaigns with custom templates (DHL, GitHub, eBay, Microsoft 365 themes) and credential-harvesting landing pages for internal security awareness training.
- Collaborated with SOC and IT teams to investigate security incidents and strengthen network defence strategies.

Intigriti Bug Bounty Platform

Remote

Independent Security Researcher

Jun 2026 - Present

- Discovered a CVSS 9.1 Critical vulnerability in a major European media publisher (50+ brands); confirmed valid by triage team.
- Identified a High-severity cross-tenant IDOR in a European HR SaaS platform via session cookie confusion on a shared subdomain domain
- Completed all 52 PortSwigger Web Security Academy Server-Side Vulnerabilities labs.

Extracurricular Activities

HackTheBox CTF

Cyprus

Participant

Dec 2024

- Ranked 241st out of 1,200 teams (top 20%) in a competitive cybersecurity challenge, demonstrating strong ethical hacking and problem-solving skills.

Introduction to Information Security Requirements - Erasmus

Sweden

Participant

Feb 2025

- Completed intensive program on system security requirements and secure network environment best practices.

Hackathon by hack{cyprus}

Cyprus

Participant

Nov 2023

- Explored AI-driven solutions to real-world environmental challenges in a competitive hackathon environment.